



Organizing A **Page's Content**

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Outline

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Unordered Lists

it is legal to omit the `</li>` end tag
for list elements

```
<ul>  
  <li>Wake up at 9ish.</li>  
  <li>Go to school.</li>  
</ul>
```

Parent and Child Elements

A red arrow points from a red text box in the upper right to a gray rectangular area in the center of the slide. The text box contains the text "Nested lists (list inside a list)".

Nested lists
(list inside a list)

Symbol for Unordered List Items

- According to the W3C, the *default symbol* for unordered list items is a **bullet** for all levels in a nested list, but the major browsers typically use **bullet**, **circle**, and **square** symbols for the *different levels* in a nested list.
- Because the official symbol defaults and the browser symbol defaults are different, you should **avoid** *relying on them*.
- Instead, you should use **CSS's list-style-type** property to *explicitly* specify the symbols used in your web page lists.

Symbol for Unordered List Items

- For unordered lists, the *most popular values* for the *list-style-type* property are **none**, **disc**, **circle**, and **square**.
- none → browser displays *no symbol* next to each list item.
- disc → generates *bullet symbols*.
- circle & square → generate *hollow circles* and filled-in *squares*, respectively

```
<style>
  ul {list-style-type: circle;}
  ul ul {list-style-type: square;}
</style>
```

Descendant Selectors

- For each pair of adjacent selectors, the browser searches for a pair of elements that match the selectors such that the second element is contained within the first element's start tag and end tag.
- When **an element** is *inside* **another element's** start tag and end tag, we say that the element is a descendant of the outer element.

```
<style>
  ul {list-style-type: disc;}
  ul ul {list-style-type: square;}
  ul ul ul {list-style-type: none;}
</style>
```

Descendant Selectors

- If you ever have **two** or **more** CSS rules that *conflict*, the **more specific** rule **wins**.
- That's a general principle of programming and you should remember it—the more specific rule wins.

Ordered Lists

- How to generate list items that have *numbers* and *letters* next to them.
- Numbers and letters indicate that the **order** of the list items is important, and that changing the order would change the list's meaning.
- Those types of lists are referred to as *ordered lists*.

```
<ol>  
  <li>Personal experiences</li>  
  <li>What friends say</li>  
</ol>
```

Figures

- Typically, a **figure** holds *text*, *programming code*, an *illustration*, a *picture*, or a *data table*. As with all figures, the figure element's content should be self-contained, and it should be referenced from elsewhere in the web page.



```
figure {  
    border: solid crimson;  
    padding: 6px;  
}
```

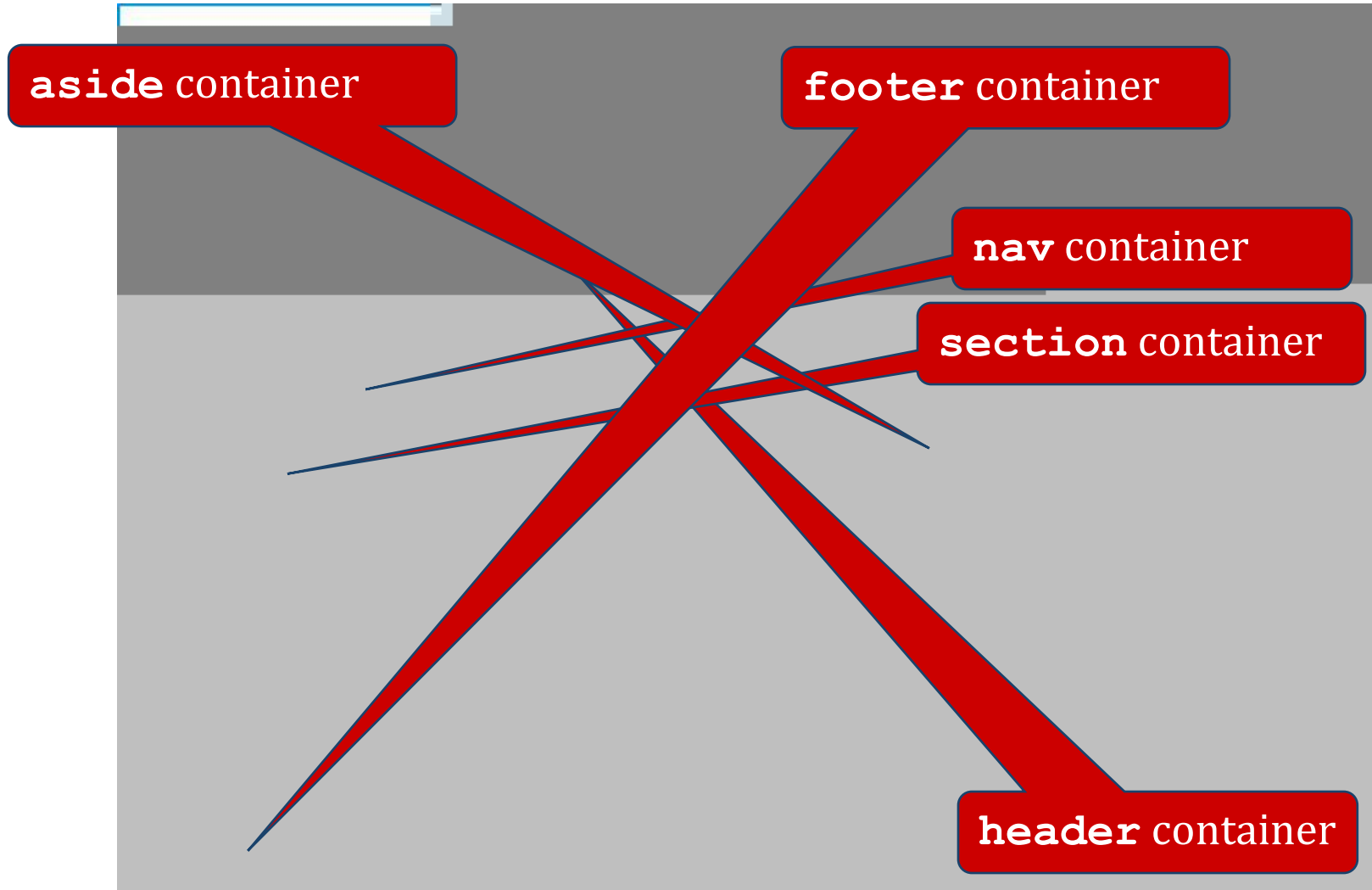
Figures

Figures

Organizational Elements

- To group web page content into sections so that we can use **CSS** and **JavaScript** to manipulate their content more effectively.
- The organizational elements:
 - **section**
 - **article**
 - **aside**
 - **nav**
 - **header**
 - **footer**

Organizational Elements



Organizational Elements

section element

- **section**: ➔ container: used to group together a section of a web page.
- Typically, a section will contain a *heading* and *one or more paragraphs*, with the *heading* saying *something about the common theme*

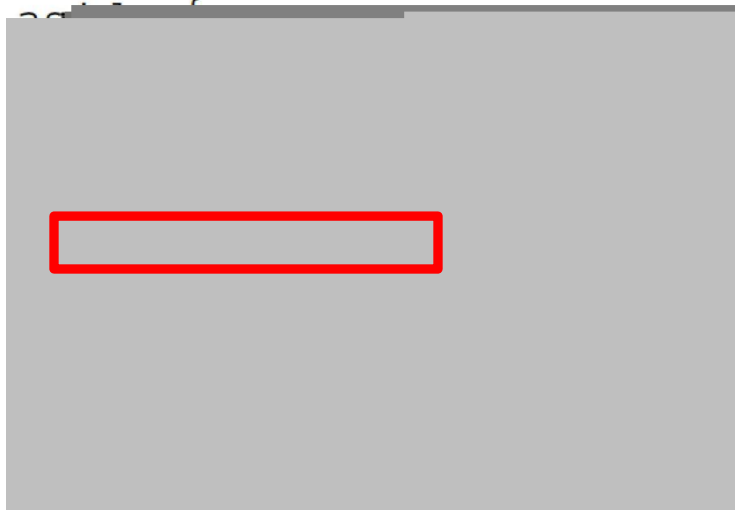
section element

article element

- **article** → grouping together *one* or *more sections* such that the *group of sections* form an *independent part* of a web page.

aside element

- **aside** → to group together *content* that has *something to do* with the rest of the web page, but it isn't part of the main flow.
- Typically, you should **position** an **aside** element at the *right* or *left*.



nav & a element

- **nav** → a container that normally contains *links* to other web pages or to other parts of the current web page
- **nav** → navigation
- **a** → implements a link



header & footer element

- **header** → for grouping together *one* or *more heading elements* (h1-h6)
- Normally, the **header** is associated with a *section*, an *article*, or the *entire web page*.
- To form that association, the **header** element must be positioned **within** its associated content container.
- Typically, that means *at the top of the container*, but it is **legal** and **sometimes** appropriate to have it positioned lower.

header & footer element



header & footer element

- **footer** → for grouping together information to form a footer.
- Typically, the **footer** holds content such as copyright data, author information, or related links.
- The **footer** should be *associated* with a **section**, an **article**, or the entire web page.
- To form that association, the **footer** element must be *positioned* within its *associated content container*.
- Typically, that means **at the bottom** *of the container*, but it is **legal** and **sometimes** appropriate to have it positioned **elsewhere**.

header & footer element



Child Selectors

- A *child selector* is a more refined version of a descendant selector.



combinator, used to
combine the elements

CSS Inheritance

- **CSS inheritance** is *when* a CSS property value *flows down* from a **parent** element to child (children)
- It parallels the inheritance of genetic characteristics (e.g., height and eye color) from a biological parent to a child.
- Some *CSS properties* are *inheritable* and some are *not*.
- CSS properties **inheritable**:



References

- W3 Schools – [w3schools.com](https://www.w3schools.com)
- Web Programming with HTML5, CSS, and JavaScript – John Dean – Jones & Bartlett Publishers, 2019.